Chan Sherwin Stephen

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Summary

Final-year Ph.D. student at NTU (expected completion Feb 2026) specializing in physics-based simulation of physical human–robot interaction (pHRI) for assistive and rehabilitative devices. Develops realistic digital human twins and personalized controllers in MuJoCo within a Real2Sim2Real workflow to study safe, intuitive HRI. Currently exploring foundation models and LLMs to improve simulation fidelity and adaptability.

Education

Nanyang Technological University, Singapore — Aug 2021 – Feb 2026 (Expected)
Ph.D. in Mechanical Engineering
- **Advisor:** Prof Ang Wei Tech
- **Proposed Dissertation:** Human-in-the-Loop Simulation for Adaptive Assistive Robots: Personalising Human Models and Robot Control
- **Research Interests:** Physics-Based Simulation, Human-Robot Interaction, Machine Learning, Foundation Models, Large Language Models

Nanyang Technological University, Singapore — Aug 2017 – May 2021
B.E. in Mechanical Engineering with a Specialization in Robotics and Mechatronics
- **GPA:** 4.84/5.00, Dean's List for 2017/2018
- **Awards:** Nanyang Scholarship

Research Projects

Human-In-The-Loop Robotic Simulator — Sept 2021 – present
A simulation framework for physical human-robot interaction (pHRI) investigation to personalise robotic systems.
- Led the development of a human-in-the-loop simulation framework for assistive robotics, featuring personalized digital twins with varying abilities and disabilities. The pipeline incorporates skeletal, musculoskeletal, and soft body models, using reinforcement learning to simulate diverse human capabilities and enable realistic, adaptive testing of robotic systems.
- Designed and validated a Real2Sim2Real framework to personalize robotic controllers for robot-assisted feeding and gait-assistive robots, enabling improved pHRI through simulation-informed adaptation and real-world testing.
- Investigated various human-robot interaction modalities in simulation, including soft body interaction dynamics, mass-spring-damper models, and other methods to accurately represent physical interactions.

Mobile Third Arm Robot — 2020 – 2022
Vision-guided mobile third arm co-bot for patient transfers between bed and wheelchair.
- Designed and developed a complete mobile third arm robot to assist caregivers in moderate assisted pivot transfers between a bed and wheelchair, covering mechanical design, electronics integration, and control systems.
- Implemented a human-tracking computer vision algorithm and developed human-robot interaction strategies to enable safe, intuitive operation between the caregiver, patient, and mobile third arm robot.

Work Experience

Project Officer, Nanyang Technological University, Singapore – Singapore — Aug 2021 – present
- Spearheaded the Human-In-The-Loop Robotic Simulator project, focusing on developing a platform that combines realistic human digital twins with assistive robots for human-in-the-loop simulations.
- Mentored seven final-year project students and collaborated with cross-functional research teams, including biomechanics experts, therapists, engineers, and research leads, to advance the development and application of the simulator.

- Prepared funding proposals ranging from small grants of SGD 200K to large multi-institutional proposals of over SGD 3 million (currently under review) to support ongoing and upcoming research initiatives.

Engineering Intern, Oishii – New Jersey, USA June 2019 – June 2020

- Designed an indoor vertical farm structure accommodating 250,000 plants with a novel rack system that increases efficiency of construction and assembly process by 90%.
- Pioneered development of a USD 400,000 R&D facility consisting of ten independent growing environments tracking more than 80 plant parameters to determine optimal plant growth and yield.
- Led an engineering team of 12 to implement testing and troubleshooting schedule to all farm sub-systems as lead test engineer to bring the world's largest indoor strawberry vertical farm facility operational.
- Evaluated 32 engineering candidates for engineering roles, mentored 19 new engineering technicians to improve mechanical and electrical skillsets and align with company working norms.

Selected Publications

ORBiT: Optimizing Robot-Assisted Bite Transfer Leveraging a Real2Sim2Real Framework Oct 2025

Sherwin Stephen Chan†, J Anne Yow†, Yi Heng San, Vasanthamaran Ravichandram, Yifan Wang, Lek Syn Lim, Wei Tech Ang

IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), † Denotes equal contribution

A Human-In-The-Loop Simulation Framework for Evaluating Control Strategies in Gait Assistive Robots May 2025

Yifan Wang†, ***Sherwin Stephen Chan†***, Mingyuan Lei, Lek Syn Lim, Henry Johan, Bingran Zuo, Wei Tech Ang

IEEE International Conference on Robotics and Automation (ICRA), † Denotes equal contribution

Personalised 3D Human Digital Twin with Soft-Body Feet for Walking Simulation Dec 2024

Kum Yew Loke, ***Sherwin Stephen Chan***, Mingyuan Lei, Henry Johan, Bingran Zuo, Wei Tech Ang

International Conference on Social Robotics (ICSR)

Creation and evaluation of human models with varied walking ability from motion capture for assistive device development Sept 2023

Sherwin Stephen Chan, Mingyuan Lei, Henry Johan, Wei Tech Ang

IEEE International Conference on Rehabilitation Robotics (ICORR)

Skills

Programming Languages: Python, C++, C

Robotics: MuJoCo, OpenSim, Reinforcement Learning, Machine Learning (PyTorch, TensorFlow), Isaac Lab (Gym), ROS, SolidWorks, SolidEdge, STM32, ESP32, Linux

Spoken Languages: English, Chinese, Hokkien (Dialect), Tagalog

Awards and Achievements

Dyson-NTU Innovation Challenge – Singapore 2021-04 – 2021-09

Won 1st place in the annual innovation challenge organized by NTUitive.

- Awarded SGD 10,000 MDT grant to prototype a innovative smart planter to address urban gardening challenges

Overseas Entrepreneurship Program (OEP) Pitch Day Award – Singapore 2020-09 – 2021-02

Founded HortiCole startup and secured funding for IoT agricultural solution.

- Awarded SGD 5,000 to develop a low-cost, modular IoT planter system with alpha prototype in 6 months
- Conducted market research and received strong market validation of 64.5% from 400 participants